

1 Flow Chart

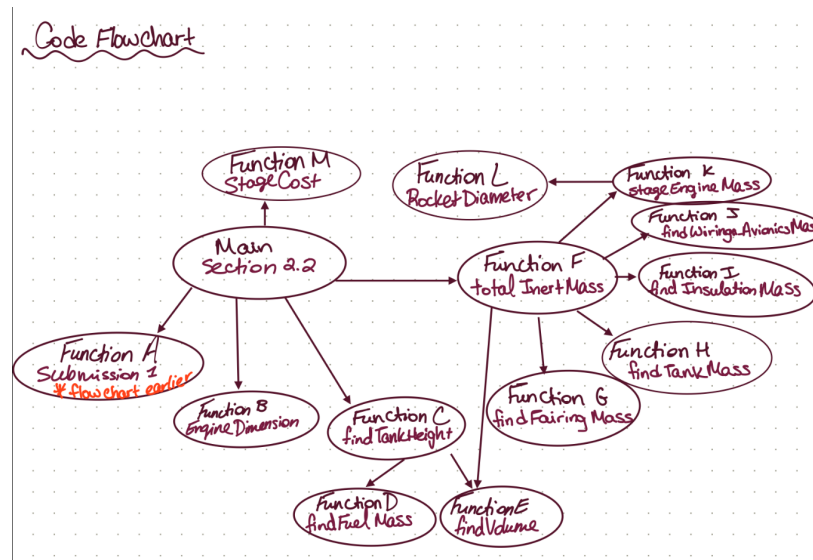


Figure 1: Flow Chart of Function Organization

2 Code

```
1 %% Main Section Code for 2.1
2 % All team members worked on this code
3
4 clear; clc; close all;
5 propNames = ["LOX/LCH4" "LOX/LH2" "LOX/RP1" "Solid" "Storables"];
6 %propNames = ["LOX/LH2"];
7 Propellantstage1 = "Storables"; % user changed
8
9 h = 4; % meters, we decided as a team vote
10 delta = 0.08;
11 %tolarence 0.05 for mass margin
12
13 vehicleParamsSize = 7; % if you add a vehicleParam later, change this number
14 allOptimizedVehicles = zeros(length(propNames)*2, vehicleParamsSize);
15 massMargins = zeros(length(propNames), 1);
16 totalMasses = zeros(2, length(propNames));
17 chiValues = zeros(length(propNames), 1);
18 deltaValues = zeros(length(propNames), 1);
19 for k = 1:length(propNames) % Going through all the propellant names/
    combinations
20
21     firstIteration = true;
22     mass_margin = 0;
23
24     while mass_margin < 0.3 || mass_margin > 0.31 % This is to update the
        delta, if it is not the first iteration, until a mass margin of 30% is
        achieved
25
26         if ~ firstIteration
27             if mass_margin < 0.3
28                 %delta = delta + 0.0001;
29                 % adaptive step
30                 delta = delta + max(abs(mass_margin - 0.3)/100, 1e-5);
31             elseif mass_margin > 0.301
32                 %delta = delta - 0.0001;
33                 % adaptive step
34                 delta = delta - max(abs(mass_margin - 0.3)/100, 1e-5);
35             end
36         else
37             firstIteration = false;
38         end
39
40         Propellants = [Propellantstage1, propNames(k)]; % user's specific
            propellant combination
41         [Mo_min, Min1_min, Min2_min, Mo1_min, Mo2_min, Mpr1_min, Mpr2_min,
            chi_min] = submission1(delta, Propellantstage1, propNames(k), 0);
            % Grab submission 1 masses
42         chiValues(k) = chi_min;
43
44         vehicle_inertMass1 = Min1_min + Min2_min;
45         m0 = [Mo_min, Mo2_min]; % stage 1 and 2 array
46         totalMasses(:, k) = m0;
```

```
47     Mpr0 = [Mpr1_min Mpr2_min]; % stage 1 and 2 array
48     thrust_weight_ratio = [1.3 0.76];
49     vehicle_inertMass2 = 0;
50     previous_radius = 0;
51     previous_stage = [];
52
53     vehicleParams = [[]; []];
54
55     for i=2:-1:1 % start with stage 2 and go to stage 1
56         stage = i;
57         [nEngines, diameter_ofthrust] = EngineDimension(stage, m0(stage),
58             Propellants(stage)); % find number of engines and diameter of
59             all those engines
60         radius = diameter_ofthrust/2;
61
62         [height, fuelh, h_oxidizer, ratio, rho] = findTankHeight(Mpr0(i),
63             radius, Propellants(stage)); % function to find height of
64             tank
65         min_radius = Inf; % intializing variable, high number
66         min_height = Inf;
67         minInertmass = Inf; % intializing variable, high number
68         minstageArray = [];
69         minEngines = Inf;
70         lastHeight = Inf;
71
72         firstRun = true;
73
74         while (radius < height*10 && height ~= lastHeight) || firstRun %
75             constraint
76                 firstRun = false;
77                 radius = radius + 0.1; % keep increasing radius until
78                 constraint is met
79
80                 lastHeight = height;
81                 [height, fuelh, h_oxidizer, ratio, rho] = findTankHeight(Mpr0(
82                     i), radius, Propellants(i)); % new height based on new
83                     radius
84                 L = height;
85                 D = radius*2;
86
87                 if L/D > 13 || radius < previous_radius || D < 5.3 % if
88                     condition not met then skip to the next iteration
89                     continue
90                 end
91
92                 %array = [1:stage height, 2:radius of stage, 3:height of fuel
93                     tank,
94                     % 4:height of oxidizer tank, 5:Propellant mass, 6:fuel
95                     fraction,
96                     % 7:oxidizer fraction, 8:fuel density, 9:oxidizer density, 10:
97                     initial mass
98                     % 11: Thrust to Weight ratio, 12:number of engines]
99                 stageArray = [height, radius, fuelh, h_oxidizer, Mpr0(stage),
100                     ratio(2), ratio(1), rho(2), rho(1), m0(stage),
```

```
        thrust_weight_ratio(stage), nEngines];
88     stage1_array = [];
89     stage2_array = stageArray;
90     if stage == 1
91         stage1_array = stageArray;
92         stage2_array = previous_stage;
93     end
94
95     [InertMass, stage, propellant_names] = totalInertMass(stage1_array
    ,stage2_array, Propellants(stage), m0(stage), h, stage);
96     currentInertMass = sum(InertMass);
97
98     if currentInertMass < minInertmass    % check for new minimum
        inert mass
99         min_radius = radius;                % new minimum radius
100        min_height = height;                % new minimum height
101        minInertmass = currentInertMass;    % new minimum inert
        mass
102        minstageArray = stageArray;
103        minEngines = nEngines;
104    end
105 end
106 previous_radius = min_radius;    % update the previous radius
107 vehicle_inertMass2 = minInertmass + vehicle_inertMass2;    % add
    the inert masses
108 previous_stage = minstageArray;
109
110 vehicleParams(stage,:) = [minEngines min_radius min_height
    minInertmass Mpr0(stage) m0(stage) thrust_weight_ratio(stage)
    ];
111
112 end
113
114 allOptimizedVehicles(2*k-1:2*k, :) = vehicleParams;
115 mass_margin = (vehicle_inertMass1 - vehicle_inertMass2)/(
    vehicle_inertMass2); % calculate the mass margin
116 massMargins(k) = mass_margin
117 deltaValues(k) = delta;
118
119 end % end of while loop
120
121 end
122
123 for k=1:(length(allOptimizedVehicles(:,1))/2)
124     disp(Propellantstage1 + ", " + propNames(k))
125     for l=0:1
126         index = 2*k+l-1;
127         fprintf(" Stage: %d \tEngines: %d \tRadius: %f \tHeight: %f \tInert Mass: %
            d \t\nPropellant Mass: %d \tInitial Stage Mass: %d \tTWR: %d \tMass
            Margin: %d\n\n", ...
128             l+1, ...
129             allOptimizedVehicles(index, 1), ...
130             allOptimizedVehicles(index, 2), ...
131             allOptimizedVehicles(index, 3), ...
```

```
132         allOptimizedVehicles(index, 4), ...
133         allOptimizedVehicles(index, 5), ...
134         allOptimizedVehicles(index, 6), ...
135         allOptimizedVehicles(index, 7), ...
136         massMargins(k))
137     end
138 end
139
140 for k=1:(length(allOptimizedVehicles(:,1))/2)
141     index = 2*k-1;
142     nEngines = [allOptimizedVehicles(index, 1) allOptimizedVehicles(index+1,
143         1)];
144     radii = [allOptimizedVehicles(index, 2) allOptimizedVehicles(index+1, 2)];
145     heights = [allOptimizedVehicles(index, 3) allOptimizedVehicles(index+1, 3)
146         ];
147
148     Mpr0 = [allOptimizedVehicles(index, 5) allOptimizedVehicles(index+1, 5)];
149     [height1, fuelh1, h_oxidizer1, ratio1, rho1] = findTankHeight(Mpr0(1),
150         radii(1), Propellantstage1); % new height based on new radius
151     [height2, fuelh2, h_oxidizer2, ratio2, rho2] = findTankHeight(Mpr0(2),
152         radii(2), propNames(k)); % new height based on new radius]
153     fuel_heights = [fuelh1 fuelh2];
154     oxidizer_heights = [h_oxidizer1 h_oxidizer2];
155     ratios = [ratio1; ratio2];
156     rhos = [rho1; rho2];
157     thrust_weight_ratio = [1.3 0.76];
158     m0 = totalMasses(:,k);
159
160     stageArray1 = [heights(1), radii(1), fuel_heights(1), oxidizer_heights(1),
161         Mpr0(1), ratios(2,1), ratios(1,1), rhos(2,1), rhos(1,1), m0(1),
162         thrust_weight_ratio(1), nEngines(1)];
163     stageArray2 = [heights(2), radii(2), fuel_heights(2), oxidizer_heights(2),
164         Mpr0(2), ratios(2,2), ratios(1,2), rhos(2,2), rhos(1,2), m0(2),
165         thrust_weight_ratio(2), nEngines(2)];
166
167     % Need: Stage, Chi, Propellant,
168     % Propellant Tanks S1, Propellant Tanks S2, Tank Insulation S1, Tank
169     % Insulation S2,
170     % Engine S1, Engine S2, Thrust Structure S1, Thrust Structure S2, Casing
171     % S1, Casing S2, Gimbals S1, Gimbals S2,
172     % Avionics (S2), Wiring S1, Wiring S2,
173     % Payload Fairing, Inter Tank Fairing S1, Inter Tank Fairing S2, Inter
174     % Stage Fairing, Aft Fairing
175
176     S2Avionics = 0;
177     inertMassSum = 0;
178     costSum = 0;
179
180     Propellants = [Propellantstage1, propNames(k)];
181
182     fprintf(" Propellants: %s %s\n\n", Propellants(1), Propellants(2))
183
184     for stage=1:2
```

```
175     [InertMass, ~, propellant_names] = totalInertMass(stage1_array ,  
176         stage2_array , Propellants(stage), m0(stage), h, stage);  
177  
178     % Array guide for subsystem masses:  
179     % 1 —> if i = 1 , propellant_name = "Solid" : totalMass1 = [Interstage  
180         Aft1 propellantTank insul_solid engineMass1 structureMass1  
181         gimbalsMass1 wiringMass1 avionicsMass1];  
182     % 2 —> if i = 1 , propellant_name = "else" : totalMass1 = [Intertank1  
183         Interstage Aft1 fuel_tank1 oxidizer_tank1 insul_oxid1 insul_fuel1  
184         engineMass1 structureMass1 gimbalsMass1 wiringMass1 avionicsMass1 ];  
185     % 3 —> if i = 2 , propellant_name = "Solid" : totalMass2 = [Payload2  
186         propellantTank2 insul_solid2 engineMass2 structureMass2 gimbalsMass2  
187         wiringMass2];  
188     % 4 —> if i = 2 , propellant_name = "else" : totalMass2 = [payload2  
189         Intertank2 fuel_tank2 oxidizer_tank2 insul_oxid2 insul_fuel2  
190         engineMass2 structureMass2 gimbalsMass2 wiringMass2];  
191  
192     displayString = "Stage: %d\tChi: %f\tPropellant: %d\tDelta: %f\n" + ...  
193         "Propellant Tanks: %d\tTank Insulation: %d\t\n" + ...  
194         "Engines: %d\tThrust Structure: %d\tCasing: %d\tGimbals: %d\t\n" + ...  
195         "Avionics: %d\tWiring: %d\n" + ...  
196         "Payload Fairing: %d\tIntertank Fairing: %d\tInterstage Fairing: %d\  
197         tAft Fairing: %d\n\n";  
198     if (stage == 1 && strcmp(propellant_names, "Solid"))  
199         S2Avionics = InertMass(9);  
200         fprintf(displayString, ...  
201             stage, chiValues(k), Mpr0(stage), deltaValues(k), ...  
202             InertMass(3), InertMass(4), ...  
203             0, InertMass(6), InertMass(5), InertMass(7), ...  
204             0, InertMass(8), ...  
205             0, 0, InertMass(1), InertMass(2));  
206     elseif (stage == 1 && strcmp(propellant_names, "else"))  
207         S2Avionics = InertMass(12);  
208         fprintf(displayString, ...  
209             stage, chiValues(k), Mpr0(stage), deltaValues(k), ...  
210             InertMass(4)+InertMass(5), InertMass(6)+InertMass(7), ...  
211             InertMass(8), InertMass(9), 0, InertMass(10), ...  
212             0, InertMass(11), ...  
213             0, InertMass(1), InertMass(2), InertMass(3));  
214     elseif (stage == 2 && strcmp(propellant_names, "Solid"))  
215         fprintf(displayString, ...  
216             stage, chiValues(k), Mpr0(stage), deltaValues(k), ...  
217             InertMass(2), InertMass(3), ...  
218             0, InertMass(5), InertMass(4), InertMass(6), ...  
219             S2Avionics, InertMass(7), ...  
220             InertMass(1), 0, 0, 0);  
221     elseif (stage == 2 && strcmp(propellant_names, "else"))  
222         fprintf(displayString, ...  
223             stage, chiValues(k), Mpr0(stage), deltaValues(k), ...  
224             InertMass(3)+InertMass(4), InertMass(5)+InertMass(6), ...  
225             InertMass(7), InertMass(8), 0, InertMass(9), ...  
226             S2Avionics, InertMass(10), ...  
227             InertMass(1), InertMass(2), 0, 0);  
228     end
```

```
219     cost = stageCost(sum(InertMass));
220     fprintf(" Stage Cost: %d\n\n", cost);
221     costSum = costSum + cost;
222 end
223     fprintf(" Vehicle Cost: %d\n\n", costSum);
224 end

1 function [nEngines, diameter_ofthrust] = EngineDimension(stage,m0,prop) % Emma
    and Joseph mainly worked on code
2     propNames = ["LOX/LCH4" "LOX/LH2" "LOX/RP1" "Solid" "Storables"]; %
        propellant names
3     i = find(propNames==prop) ;
4     g = 9.81;
5     TWR = [1.3 0.76]; % Thrust to weight ratio
6     Treq = [2.26 1.86 1.92 4.5 1.75; 0.745 0.99 0.61 2.94 0.67]; % Thrusts
7     nEngines = (Treq(stage,i)*10^6/(m0*g*TWR(stage)))^-1;
8     nEngines = ceil(nEngines); % Round engine count up to whole integer
9     D_engine = [2.4 2.4 3.7 3.3 1.5; 1.5 2.15 0.92 2.34 1.13]; % Engine
        diamaters
10    diameter = D_engine(stage, i);
11    [radius_ofthrust] = RocketDiameter (nEngines,diameter);
12    diameter_ofthrust = radius_ofthrust*2;
13 end

1 function [Payload, Intertank1 ,Intertank2 , Interstage , Aft] = findFairingMass
    (rs1, rs2,h)
2 %% Call this function twice for each stage
3 % Abdullah mainly worked on this code
4 %rs1: Radius of 1st stage
5 %rs2: Radius of 2nd stage
6 %h: height of aerodynamic cone
7
8 A_cone = pi*rs2*sqrt(rs2^2 + h^2); %Area for the aerodynamic fairing
9 A_cylinder_payload = 2*pi*rs2*(13); %Area for payload cylinder
10 A_payload = A_cone + A_cylinder_payload; %Total area of payload fairing
11 A_frustrum = pi*(rs1 + rs2)*sqrt((rs1-rs2)^2+(rs1+rs2+3)^2);%Area for inter-
    stage fairing
12
13 % leave sufficient engine fairing below propellant tanks (3m)
14 A_cylinder1 = 2*pi*rs1*(2*rs1); %Area for inter-tank fairing, stage 1
15 A_cylinder2 = 2*pi*rs2*(2*rs2); %Area for inter-tank fairing, stage 2
16 A_aft = 2*pi*rs1*(rs1+3); %Area for aft fairing
17
18 %caluclate fairing masses
19 Payload = 4.95*(A_payload)^1.15;%Only for 2nd stage
20 Intertank1 = 4.95*(A_cylinder1)^1.15; %Inter-tank fairng for 1st stage
21 Intertank2 = 4.95*(A_cylinder2)^1.15; %Inter-tank fairng 2nd stage
22 Interstage = 4.95*(A_frustrum)^1.15; %One inter-stage fairing
23 Aft = 4.95*(A_aft)^1.15;%Aft fairing of 2nd stage = Inter-stage fairing
24 end

1 %% Oxidizer/fuel Mass finder
2 % Abdullah mainly worked on this code
3 function [oxidizer_mass, fuel_mass] = findFuelMass(Mpr0, ratio_array)
```

```
4      %Mpr0 = total propellant mass per stage
5      %ratio_array(1) = fuel fraction
6      %ratio_array(2) = oxidizer fraction
7      tot_ratio = ratio_array(1) + ratio_array(2);
8      oxz_ratio = ratio_array(1)/tot_ratio;
9      ful_ratio = ratio_array(2)/tot_ratio;
10     oxidizer_mass = oxz_ratio*Mpr0;
11     fuel_mass = ful_ratio*Mpr0;
12 end

1 %% Find Tank Insulation
2 % Anderson and Mahek mainly worked on this code
3 function InsMass = findInsulationMass(AT,fuelname) %exposed surface area of
    tank, fuel name
4     if strcmp(fuelname, "LH2")
5         InsMass = 2.88*AT;
6     elseif strcmp(fuelname, "LCH4") || strcmp(fuelname, "LOX")
7         InsMass = 1.123*AT;
8     else
9         InsMass = 0;
10    end
11 end

1 function [h_total, fuel_height, oxidizer_height, ratio, rho, r] =
    findTankHeight(Mpr0,r, Propellant)
2 % Abdullah Mahek Anderson mainly worked on this code
3 %Mpr0: Overall Propellant mass (kg)
4 %r:
5     % Find height of the tank cylinder based on propellant mass and radius
6 density = [71 1140 820 423 1680 1442 791]; % LH2 LOX RP-1 LCH4 APCP(solid)
    N2O4 UDMH
7 % NOTE: propMass must include oxidizer first
8 if Propellant == "LOX/LCH4"
9     rho = [density(2) density(4)];
10    ratio = [3.6 1];
11    [oxidizer_mass, fuel_mass] = findFuelMass(Mpr0, ratio);
12    propMass = [oxidizer_mass, fuel_mass];
13 elseif Propellant == "LOX/LH2"
14    rho = [density(2) density(1)];
15    ratio = [6.03 1];
16    [oxidizer_mass, fuel_mass] = findFuelMass(Mpr0, ratio);
17    propMass = [oxidizer_mass, fuel_mass];
18 elseif Propellant == "LOX/RP1"
19    rho = [density(2) density(3)];
20    ratio = [2.72 1];
21    [oxidizer_mass, fuel_mass] = findFuelMass(Mpr0, ratio);
22    propMass = [oxidizer_mass, fuel_mass];
23 elseif Propellant == "Solid"
24    ratio = [1 0];
25    rho = [density(5) 0];
26    propMass = Mpr0;
27 elseif Propellant == "Storables"
28    rho = [density(6) density(7)];
29    ratio = [2.67 1];
```



```
30     [oxidizer_mass, fuel_mass] = findFuelMass(Mpr0, ratio);
31     propMass = [oxidizer_mass, fuel_mass];
32 end
33
34 h_total = 0;
35 heights = [];
36 for i = 1:length(propMass)
37     volume = findVolume(propMass(i), rho(i));
38     %volume for cylinder with hemispherical caps
39     %Replace formula below
40     if Propellant == "Solid"
41         h = volume/(pi*r^2);
42         r = 0;
43     else
44         h = max((volume-4*(pi/3)*r^3) / (pi*r^2), 0);
45         if h == 0
46             r = (3*volume/(4*pi))^(1/3);
47         end
48     end
49
50     heights(end+1) = h+2*r;
51     h_total = h+h_total+2*r;
52 end
53 if Propellant == "Solid"
54     ratio = [0 1];
55     rho = [0 density(5)];
56 end
57 if (length(heights) == 2)
58     oxidizer_height = heights(1);
59     fuel_height = heights(2);
60 elseif (length(heights) == 1)
61     fuel_height = heights(1);
62     oxidizer_height = 0;
63 end
64 end
65
66 %% FIND Mass of the Tank
67 % Abdullah and Joseph mainly worked on this code
68 function M_tank = findTankMass(Mpr, fuelname, Vpr)
69     if strcmp(fuelname, "LOX") %LOX
70         M_tank = 0.0107*Mpr;
71     elseif strcmp(fuelname, "LH2") %LH2
72         M_tank = 0.128*Mpr;
73     elseif strcmp(fuelname, "RP1") %RP-1
74         M_tank = 0.0148*Mpr;
75     elseif strcmp(fuelname, "LCH4") %LCH4
76         M_tank = 0.0287*Mpr;
77     else %N2O4, UDMH, Solid
78         M_tank = 12.16*Vpr;
79     end
80 end
81
82 %% Volume finder
83 % Mahek worked on this code
```

```
3 function volume = findVolume(m,rho)
4     volume = m/rho;
5 end

1 %% Find wiring and avionics masses
2 % Abdullah and Anderson mainly worked on this code
3 function [wirMass,AvcMass] = findWiringa_AvionicsMass(M0,l)
4     AvcMass = 10*(M0^0.361);
5     wirMass = 1.058*sqrt(M0)*(l^0.25);
6 end

1 function [radius_ofthrust] = RocketDiameter(nEngines,diameter)
2 % Emma and Joseph mainly worked on this code
3 radius = diameter/2;
4 % curve fit of first 20 known solutions for packing a circle in a circle
5 a = 1.486;
6 b = 0.4277;
7 radius_multiplier = a*nEngines^b;
8 radius_ofthrust = radius_multiplier*radius;
9 end

1 function [engineMass, structureMass, gimbalsMass] = stageEngineMass(m0, mProp,
    TWR, type, nEngine, stageNum)
2 % Joseph mainly worked on this code
3     %m0: total mass per stage (kg)
4     %mprop: propellant mass per stage (kg)
5     %TWR: thrust to weight ratio
6     %Type: propellant name (fuel/oxidizer)
7     %nEngine: number of engines
8     %stageNum: stage number
9     g = 9.81;
10    propNames = { 'LOX/LCH4', 'LOX/LH2', 'LOX/RP1', 'Solid', 'Storables' };
11    % arrays of thrust, chamber pressure, and expansion ratios per stage per
12    % propellant
13    s1Thrust = [2.26e+6, 1.86e+6, 1.92e+6, 4.5e+6, 1.75e+6];
14    s2Thrust = [0.745e+6, 0.099e+6, 0.061e+6, 2.94e+6, 0.067e+6];
15    s1Pressure = [35.16e+6, 20.64e+6, 25.8e+6, 10.5e+6, 15.7e+6];
16    s2Pressure = [10.1e+6, 4.2e+6, 6.77e+6, 5e+6, 14.7e+6];
17    s1ExpantionRatio = [34.34, 78, 37, 16, 26.2];
18    s2ExpantionRatio = [45, 84, 14.5, 56, 81.3];
19    %Loop through propellant names and find values based on propellant pick
20    for i = 1:length(propNames)
21        if strcmp(type, propNames(i))
22            if stageNum == 1
23                TPerEngine = s1Thrust(i);
24                P0 = s1Pressure(i);
25                expanRatio = s1ExpantionRatio(i);
26            end
27            if stageNum == 2
28                TPerEngine = s2Thrust(2);
29                P0 = s2Pressure(i);
30                expanRatio = s2ExpantionRatio(i);
31            end
32        end
    end
end
```

```
33     end
34     T_total = TWR * m0 * g;%Total thrust (N)
35     structureMass = 2.55*10^-4*T_total;%overall structure mass
36     gimbalsMass = nEngine*(237.8*(TPerEngine/P0)^0.9375);%Gimbal mass per
        engine
37
38     if strcmp(type, "LOX/LH2") || strcmp(type, "LOX/LCH4") || strcmp(type, "
        LOX/RP1") || strcmp(type, "Storables")
39         engineMass = 7.81*10^-4*TPerEngine + 3.37*10^-5*TPerEngine*sqrt(
            expanRatio) + 59;% Engine mass
40         engineMass = nEngine*engineMass;
41     end
42     if strcmp(type, "Solid")
43         engineMass = 0.135*mProp;
44     end
45
46 end

1 function [ Mo_min, Min1_min, Min2_min, Mo1_min, Mo2_min, Mpr1_min, Mpr2_min,
        chi_min] = submission1 (delta, Propellantstage1, Propellantstage2, newChi)
2 % Joseph and Emma mainly worked on this code
3 % Givens
4 delta_v = 12.3; % km/s
5 m_pl = 26000; % kg
6 chi = 0.2:0.01:0.8; % array
7 Isp = [327, 366, 311, 269, 285];
8 propNames = ["LOX/LCH4" "LOX/LH2" "LOX/RP1" "Solid" "Storables"];
9 for i = 1:length(propNames) %run through all propellant names
10     %select Isp based on prop. name and stage
11     if strcmp(Propellantstage1, propNames(i))
12         Isp1 = Isp(i);
13     end
14     if strcmp(Propellantstage2, propNames(i))
15         Isp2 = Isp(i);
16     end
17 end
18
19 if (newChi == 0)
20     [M01_array, M02_array, chi_array] = getMass(delta_v, m_pl, delta, chi, Isp1,
        Isp2); %getting stage mass for all chi values
21     [m_pr1, m_pr2] = propMass(delta, M01_array, M02_array, m_pl); %propellant
        mass for both stages
22     [m_in1, m_in2] = inertMass(delta, M01_array, M02_array); %inert mass for
        both stages
23     Mo = M01_array + M02_array; %total initial mass
24     %calculate stage cost based on inert mass
25
26     costS1 = stageCost(m_in1);
27     costS2 = stageCost(m_in2);
28     costTotal = costS1 + costS2;
29     [~, cIndex] = findMinCost(costTotal); %finding the minimum of the costs
30
31     %getting all the minimum cost points for the different mass parameters
32     Mo_min = Mo(cIndex);
```

```
33     Min1_min = m_in1(cIndex);
34     Min2_min = m_in2(cIndex);
35     Mo1_min = M01_array(cIndex);
36     Mo2_min = M02_array(cIndex);
37     Mpr1_min = m_pr1(cIndex);
38     Mpr2_min = m_pr2(cIndex);
39     chi_min = chi_array(cIndex);
40     else
41     [M01_array, M02_array, chi_array] = getMass(delta_v, m_pl, delta, [newChi],
         Isp1, Isp2); %getting stage mass for all chi values
42     [m_pr1, m_pr2] = propMass(delta, M01_array, M02_array, m_pl); %propellant
         mass for both stages
43     [m_in1, m_in2] = inertMass(delta, M01_array, M02_array); %inert mass for
         both stages
44     Mo = M01_array + M02_array; %total initial mass
45
46     %getting all the minimum cost points for the different mass parameters
47     Mo_min = Mo(1);
48     Min1_min = m_in1(1);
49     Min2_min = m_in2(1);
50     Mo1_min = M01_array(1);
51     Mo2_min = M02_array(1);
52     Mpr1_min = m_pr1(1);
53     Mpr2_min = m_pr2(1);
54     chi_min = chi_array(1);
55     end
56 end

1 %% Array of givens
2 % All team members worked on this code
3 %array = [1:stage height, 2:radius of stage, 3:height of fuel tank,
4 % 4:height of oxidizer tank, 5:Propellant mass, 6:fuel fraction,
5 % 7:oxidizer fraction, 8:fuel density, 9:oxidizer density, 10:initial mass
6 % 11: Thrust to Weight ratio, 12:number of engines]
7 %% Seperate solid fuel from other propellants in calculations
8
9 %Names = [1:stage 1 fuel name, 2:stage 1 oxidizer name, 3:stage 1 propellant
         name
10 %         4:stage 2 fuel name, 5:stage 2 oxidizer name, 6:stage 2 propellant
         name]
11
12 function [InertMass, i, propellant_names]=totalInertMass(stage1, stage2,
         propellantname, M0, h, i)
13     big_names_array = ["LOX", "LCH4", "LOX/LCH4"; "LOX", "LH2", "LOX/LH2"; "LOX
         ", "RP1", "LOX/RP1"; "Solid", "", "Solid"; "Storables", "", "Storables
         "];
14     startingIndex = 1;
15     if i == 2
16         startingIndex = 4;
17     end
18     names = ["", "", "", "", "", "", ""];
19     names(startingIndex : startingIndex+2) = big_names_array(big_names_array
         (:,3) == propellantname,:);
20 %% Stage one
```

```
21  if i == 1
22      if strcmp(names(3),"Solid")
23          propellant_names = "Solid";
24          %Fairing for solid propellants on first stage is Aft & Interstage
                fairing
25          [~, ~, ~, Interstage, Aft1] = findFairingMass(stage1(2),stage2(2),h);
26          %Propellant mass = solid fuel
27          solidM = stage1(5);
28          %DENSITY OF SOLID PROPELLANT SHOULD BE STORED IN FUEL DENSITY
29          solidV = findVolume(solidM,stage1(8));
30          propellantTank = findTankMass(solidM, names(3),solidV);
31          %find Surface area of tanks
32          solidA = 2*pi*stage1(2)*stage1(1)+4*pi*stage1(2)^2;
33          %find insulation mass
34          insul_solid = findInsulationMass(solidA, names(3));
35          %find Engine, Casing & Gimbal masses
36          [engineMass1, structureMass1, gimbalsMass1] = stageEngineMass(stage1
                (10), stage1(5), stage1(11), names(3), stage1(12), 1);
37          [~, avionicsMass1] = findWiring_AvionicsMass(M0,stage2(1));
38          [wiringMass1, ~] = findWiring_AvionicsMass(M0,stage1(1));
39          % totalMass1 = Interstage+Aft1+propellantTank+insul_solid+engineMass1+
structureMass1+gimbalsMass1+wiringMass1+avionicsMass1;
40          totalMass1 = [Interstage Aft1 0 insul_solid engineMass1 structureMass1
                gimbalsMass1 wiringMass1 avionicsMass1];
41      else
42          propellant_names = "else";
43          [~, Intertank1, ~, Interstage, Aft1] = findFairingMass(stage1(2),
                stage2(2),h);
44          %Fuel mass & Oxidizer masses
45          %Pass propellant mass and ratio
46          [Moxidizer1, Mfuel1] = findFuelMass(stage1(5), [stage1(6), stage1(7)])
                ;
47          %Find volume of fuel & oxidizer
48          V_fuel1 = findVolume(Mfuel1,stage1(8));%Pass fuel density
49          V_oxid1 = findVolume(Moxidizer1,stage1(9));%Pass oxidizer density
50          %Find tanks masses
51          fuel_tank1 = findTankMass(Mfuel1, names(1), V_fuel1);
52          oxidizer_tank1 = findTankMass(Moxidizer1, names(2), V_oxid1);
53          %find Surface area of tanks
54          Asurf_fuel1 = 2*pi*stage1(2)*stage1(3) +4*pi*stage1(2)^2;%surface area
                of fuel tank
55          Asurf_oxidizer1 = 2*pi*stage1(2)*stage1(4) +4*pi*stage1(2)^2;%surface
                area of oxidizer tank
56          insul_fuel1 = findInsulationMass(Asurf_fuel1, names(1)); %insulaiton
                mass for fuel tank
57          insul_oxid1 = findInsulationMass(Asurf_oxidizer1, names(2)); %
                insulation mass for oxidizer tank
58          % find Engine, Casing, & gimbal masses
59          [engineMass1, structureMass1, gimbalsMass1] = stageEngineMass(stage1
                (10), stage1(5), stage1(11), names(3), stage1(12), 1);
60          %find Wiring mass
61          [~, avionicsMass1] = findWiring_AvionicsMass(M0,stage2(1));
62          [wiringMass1, ~] = findWiring_AvionicsMass(M0,stage1(1));
63          totalMass1 = [Intertank1 Interstage Aft1 fuel_tank1 oxidizer_tank1
```

```
        insul_oxid1 insul_fuel1 engineMass1 structureMass1 gimbalsMass1
        wiringMass1 avionicsMass1 ];

64     end
65     InertMass = totalMass1;
66 end
67 %% Stage two
68 if i == 2
69     if strcmp(names(6),"Solid")
70         propellant_names = "Solid";
71         %Fairing for solid propellants on second stage is payload fairing
72         [Payload2, ~, ~, ~, ~] = findFairingMass(0,stage2(2),h);
73         %Propellant mass = solid fuel
74         solidM2 = stage2(5);
75         %DENSITY OF SOLID PROPELLANT SHOULD BE STORED IN FUEL DENSITY
76         solidV2 = findVolume(solidM2,stage2(8));
77         propellantTank2 = findTankMass(solidM2, names(6),solidV2);
78         %find Surface area of tanks
79         solidA2 = 2*pi*stage2(2)*stage2(1) +4*pi*stage2(2)^2;
80         %find insulation mass
81         insul_solid2 = findInsulationMass(solidA2,names(6));
82         %find Engine, Casing & Gimbal masses
83         [engineMass2, structureMass2, gimbalsMass2] = stageEngineMass(stage2
            (10), stage2(5), stage2(11), names(6), stage2(12), 2);
84
85         [wiringMass2,~] = findWiringAvionicsMass(stage2(10),(stage2(1)+13+h)
            );
86         %height = 2nd stage height + payload fairing height
87         totalMass2 = [Payload2 0 insul_solid2 engineMass2 structureMass2
            gimbalsMass2 wiringMass2];
88     else
89         propellant_names = "else";
90         %Fairing mass
91         [payload2, ~, Intertank2, ~, ~] = findFairingMass(0,stage2(2),h);
92         %Fuel mass & Oxidizer masses
93         %Pass propellant mass and ratio
94         [Moxidizer2, Mfuel2] = findFuelMass(stage2(5), [stage2(6), stage2(7)])
            ;
95         %Find volume of fuel & oxidizer
96         V_fuel2 = findVolume(Mfuel2,stage2(8));%Pass fuel density
97         V_oxid2 = findVolume(Moxidizer2,stage2(9));%Pass oxidizer density
98         %Find tanks masses
99         fuel_tank2 = findTankMass(Mfuel2,names(4),V_fuel2);
100        oxidizer_tank2 = findTankMass(Moxidizer2,names(5),V_oxid2);
101        %find Surface area of tanks
102        Asurf_fuel2 = 2*pi*stage2(2)*stage2(3) +4*pi*stage2(2)^2;%surface area
            of fuel tank
103        Asurf_oxidizer2 = 2*pi*stage2(2)*stage2(4) +4*pi*stage2(2)^2;%surface
            area of oxidizer tank
104        insul_fuel2 = findInsulationMass(Asurf_fuel2,names(4)); %insulation
            mass for fuel tank
105        insul_oxid2 = findInsulationMass(Asurf_oxidizer2,names(5)); %
            insulation mass for oxidizer tank
106        % find Engine, Casing, & gimbal masses
107        [engineMass2, structureMass2, gimbalsMass2] = stageEngineMass(stage2
```

```

    (10), stage2(5), stage2(11), names(6), stage2(12), 2);
108 %find Wiring mass and avionics
109 [wiringMass2, ~] = findWiringAvionicsMass(stage2(10), (stage2(1)+13+h)
    );
110 totalMass2 = [payload2 Intertank2 fuel_tank2 oxidizer_tank2
    insul_oxid2 insul_fuel2 engineMass2 structureMass2 gimbalsMass2
    wiringMass2];
111
112 end
113 InertMass = totalMass2;
114 end
115
116 end
```